

MORPHOLOGY OF PERMANENT MANDIBULAR INCISORS

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OUTLINE OF LECTURE



Class Traits



Chronology
and
Dimensions



Morphology of
Central Incisor



Morphology of
Lateral Incisor



Differences
and
Comparison
s



Questions

CONTRASTS B/W MAXILLARY AND MANDIBULAR INCISORS

MAXILLARY INCISORS

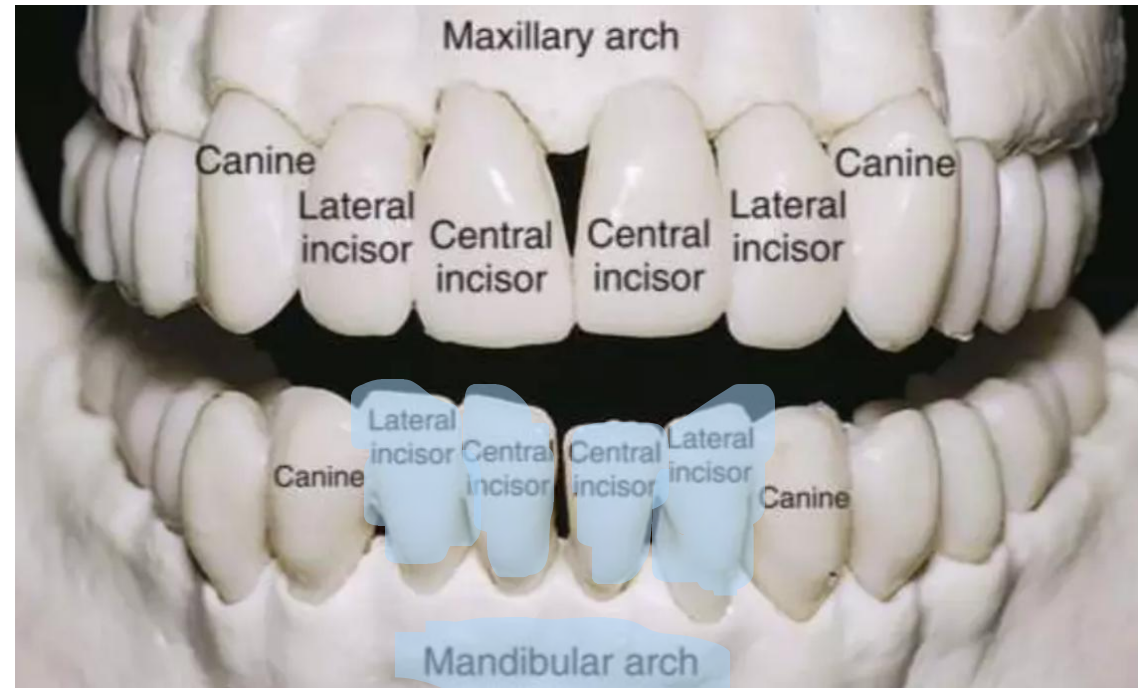
- Less symmetrical, wider crowns mesiodistally
- More rounded mesial and distal incisal angles
- Contact areas more cervical
- Pronounced marginal ridges, lingual & larger cingulum, lingual pits
- Incisal edge is labial to mid-root axis line
- Crowns wider mesiodistally than faciolingually
- No root depressions

MANDIBULAR INCISORS

- More symmetrical, narrower crowns mesiodistally
- Squarer mesial and distal incisal angles
- Contact areas are very incisal
- Muted marginal ridges, shallower lingual fossa, smaller cingulum & no pits
- Incisal edge is lingual to mid-root axis line
- Crowns are wider faciolingually than mesiodistally
- Root depressions present

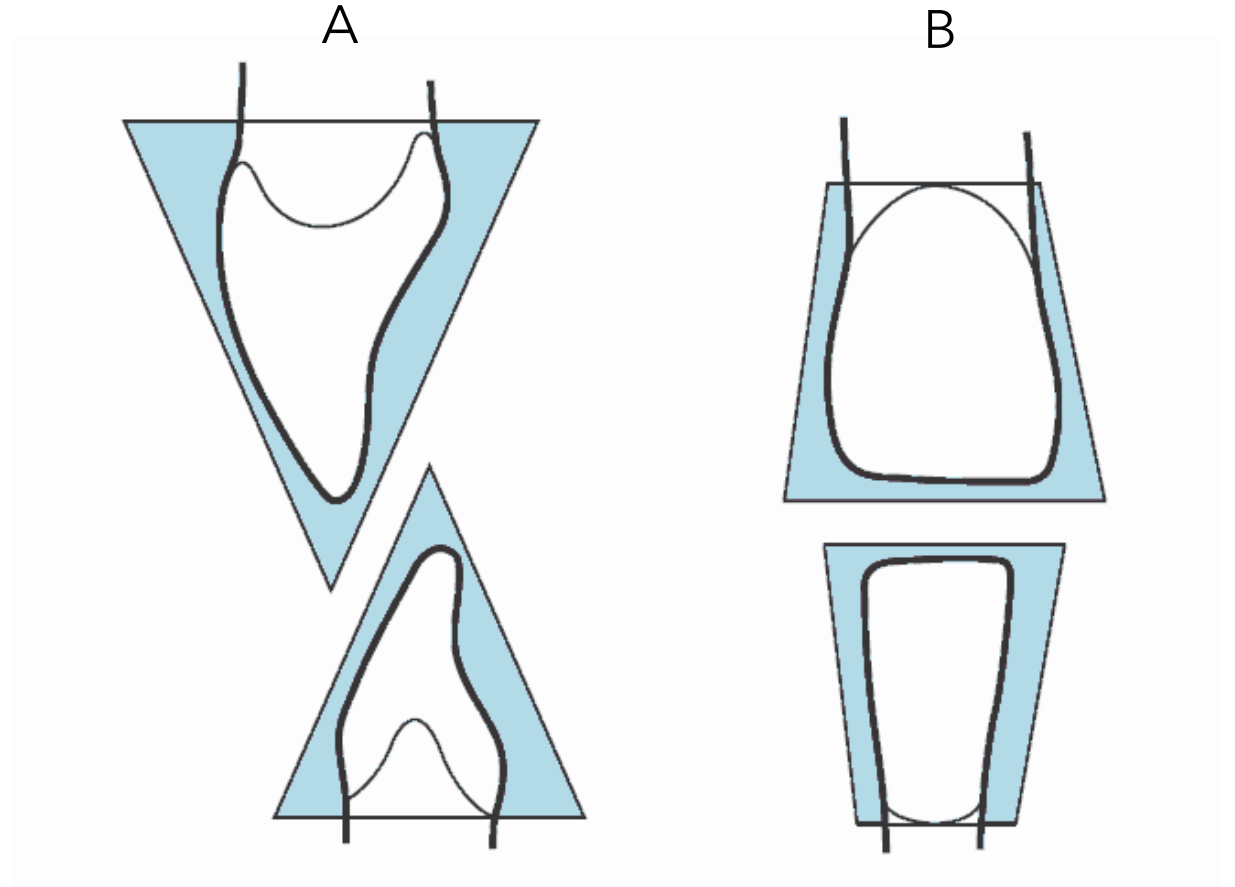
CLASS TRAITS

- The facial surface forms from three labial lobes: mesial, middle, distal lobe and lingual.
- The incisal 2/3rd of the crowns appears flattened.
- Distinct and rounded protuberances termed mamelons represent developmental lobes surmount the incisal margins of newly erupted incisors
- The marginal ridges of all incisors are located on the mesial and distal borders of the lingual surfaces



GEOMETRIC OUTLINES OF INCISORS

A: mesial or distal: triangle
B: labial or lingual: trapezoid



MANDIBULAR INCISORS

Mandibular central incisor:

- ✓ **Smallest tooth in the oral cavity (all dimensions)**
- ✓ **bilaterally symmetrical**

Mandibular lateral incisor:

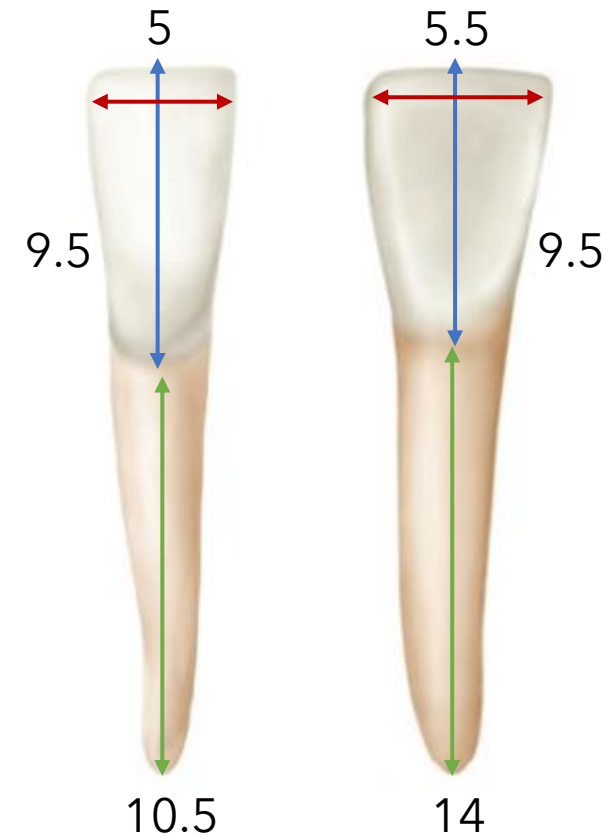
- ✓ **Larger in dimensions than mandibular central incisor**
- ✓ **Distal twist in the incisal edge**

CHRONOLOGY OF MANDIBULAR INCISORS

	Initiation of Calcification	Crown Completion	Eruption	Root Completion
Central Incisor	3-4 months	4-5 years	6-7 years	9 years
Lateral Incisor	3-4 months	4-5 years	7-8 years	10 years

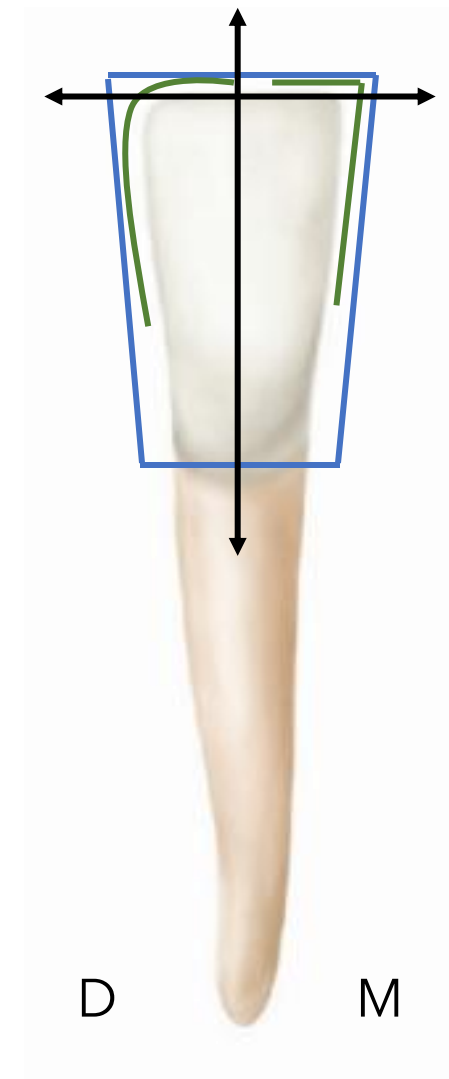
DIMENSIONS OF MANDIBULAR INCISORS

DIMENSIONS	MAND CENTRAL	MAND LATERAL
Cervico-incisal length of crown	9.5 mm	9.5 mm
Length of root	10.5 mm	14 mm
Mesiodistal diameter of crown	5 mm	5.5 mm
Labiolingual diameter of crown	6 mm	6.5 mm



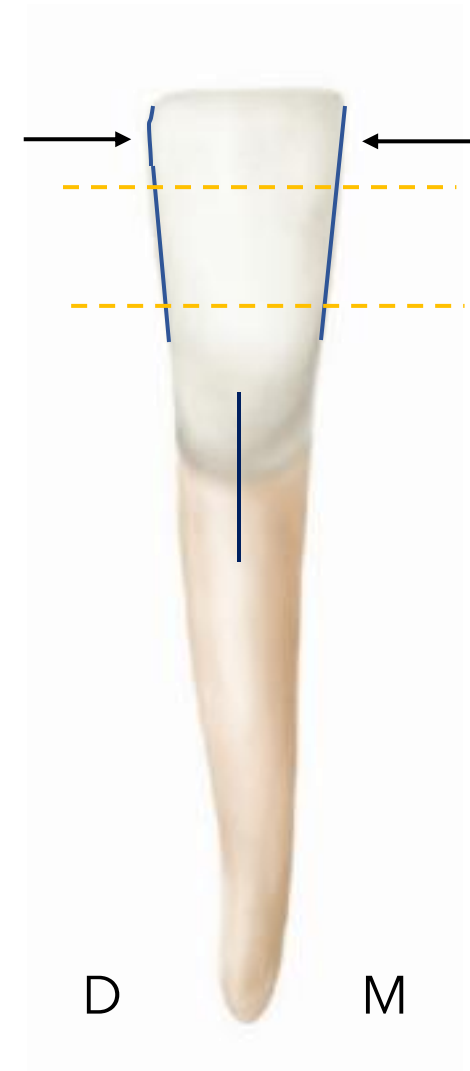
MANDIBULAR CENTRAL INCISOR LABIAL VIEW

- Crown outline trapezoidal with longer parallel side at incisal edge
- The incisal ridge of the crown is straight and is at approximately a right angle to the long axis
- Mesial and distal incisal line angles are both sharp; the distal incisal line angle is slightly more rounded.



MANDIBULAR CENTRAL INCISOR LABIAL VIEW

- The mesial and distal contact areas are at the same level (incisal 1/3rd of the crown)
- The mesial outline may be slightly straighter than the distal outline
- The curve of the cervical line is slightly to the distal of the mesiodistal long axis bisector



MANDIBULAR CENTRAL INCISOR LABIAL VIEW

- The root is narrow and converges evenly to a relatively pointed apex
- The apical third of the root terminates in a small, pointed taper in most cases curving distally.



MANDIBULAR CENTRAL INCISOR

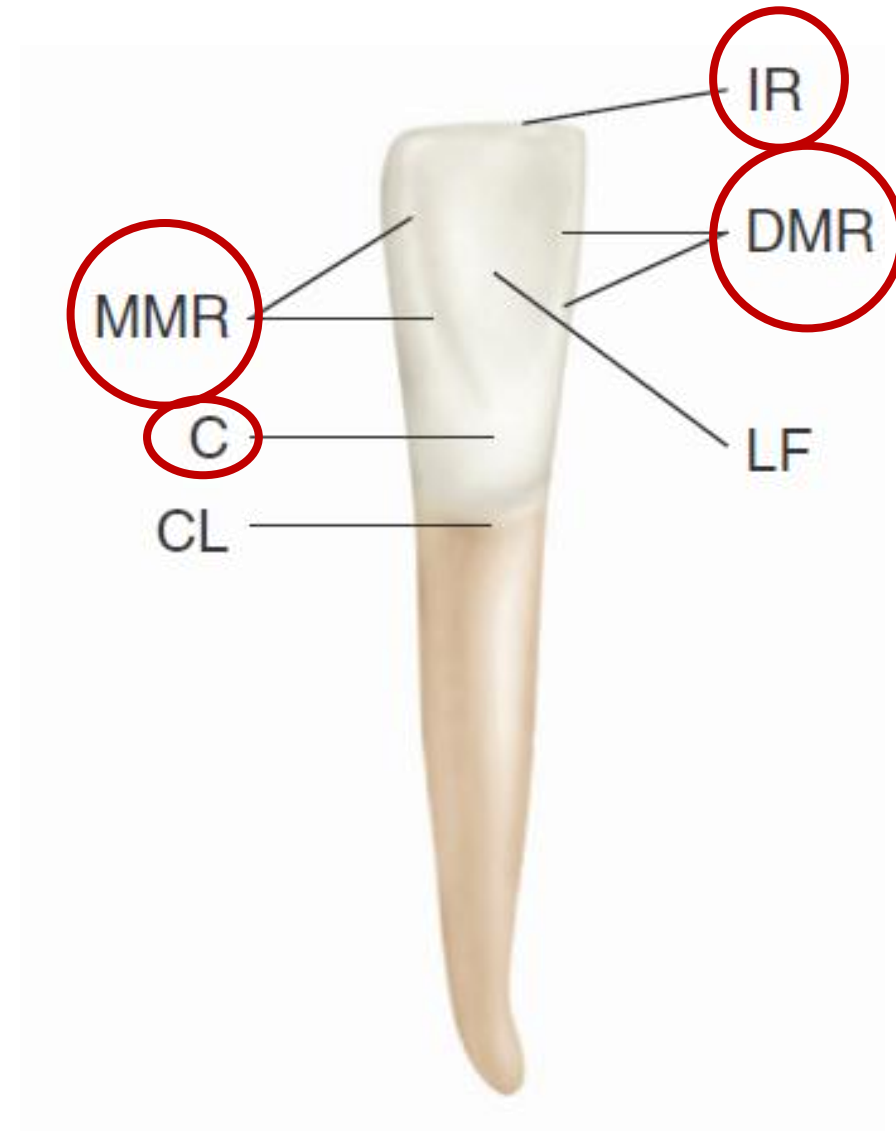
LINGUAL VIEW

- The anatomical features found on the lingual surfaces are present but relatively muted
- The lingual surface has following features:
 - -Lingual fossa: shallow concavity/trapezoidal
 - -Cingulum: smooth, rounded convexity
 - -Marginal ridges: linear and muted



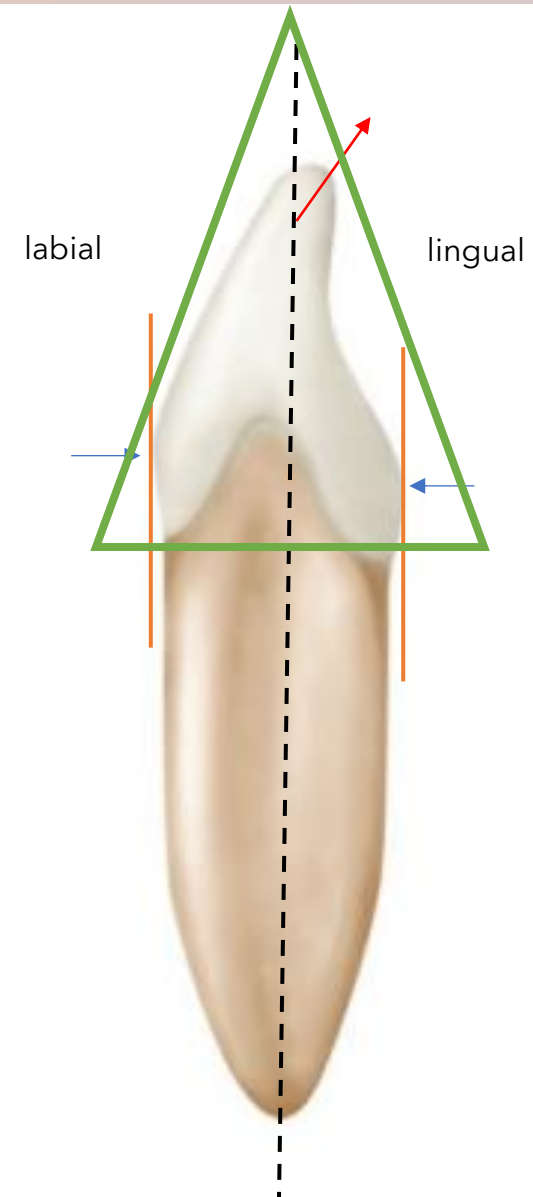
■ **Boundaries of the lingual fossa**

- Mesially: mesial marginal ridge
- Distally: distal marginal ridge
- Incisally: Lingual portion of incisal ridge
- Cervically: cingulum



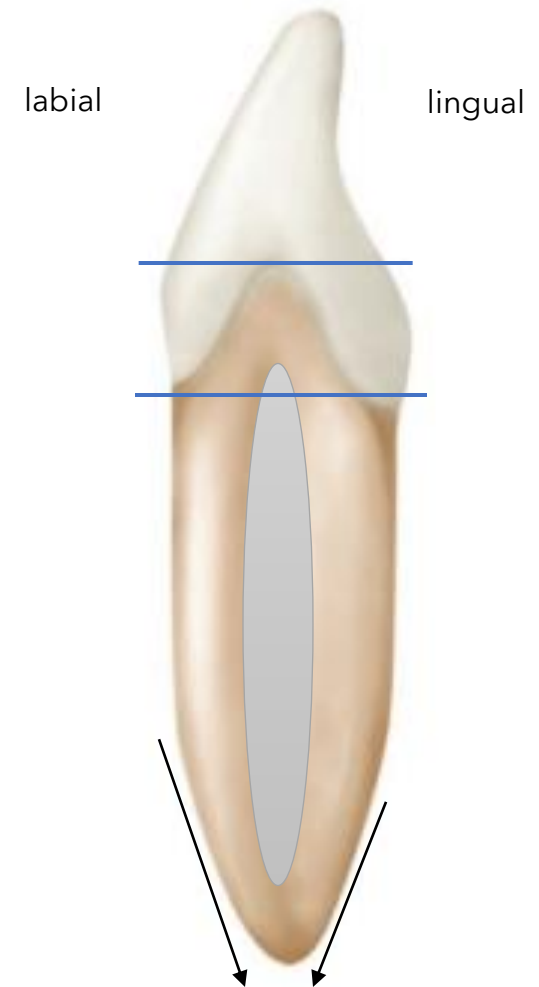
MANDIBULAR CENTRAL INCISOR MESIAL VIEW

- The crown outline is wedge-shaped
- The incisal edge will be lingual to the facio-lingual long axis bisector (center of root)
- The crests of the cervical ridges (height of contour) will not be at the same level facially and lingually



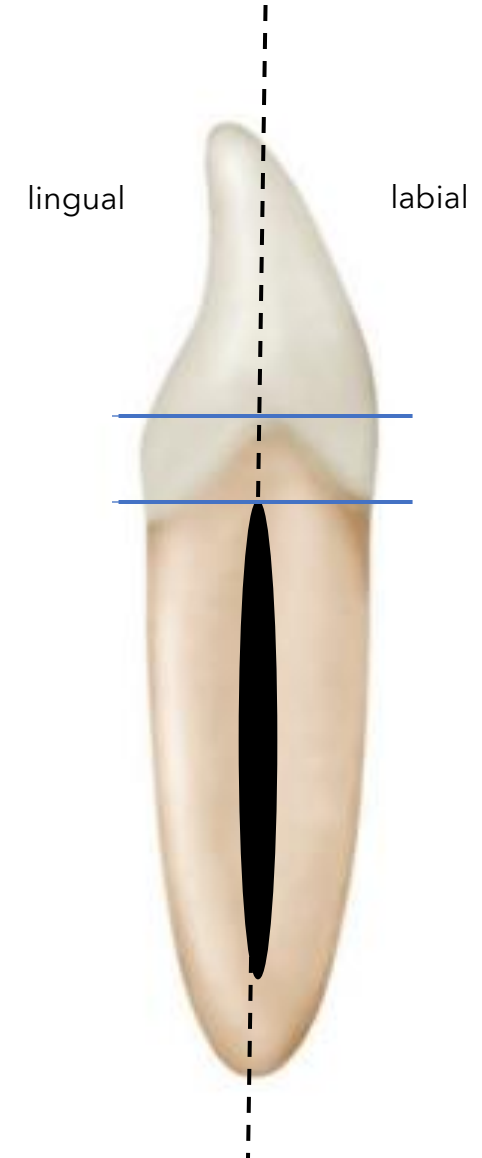
MANDIBULAR CENTRAL INCISOR MESIAL VIEW

- Lingual outline: smooth convexity of the cingulum; and a concavity of lingual fossa
- The cervical line curves more incisally
- The root tapers rapidly in the apical third to a blunt rounded end
- A broad longitudinal root depression is usually found for most of the length



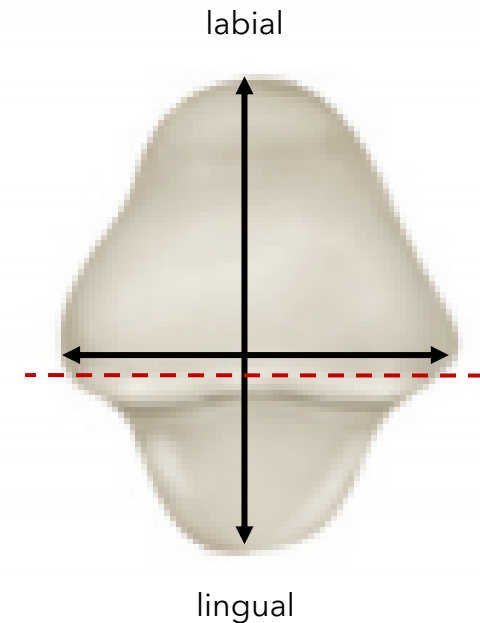
MANDIBULAR CENTRAL INCISOR DISTAL VIEW

- The cervical line curves incisally for a relatively shorter distance
- A developmental depression is more marked, with a deeper, more well-defined **developmental groove** at its center
- The apex of the root will be almost even with the labiolingual long axis bisector



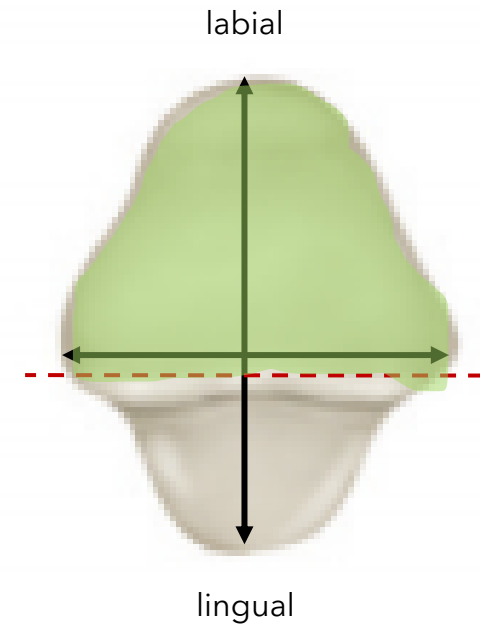
MANDIBULAR CENTRAL INCISOR INCISAL VIEW

- The incisal aspect illustrates **bilateral symmetry**. The mesial half of the crown is identical to the distal half
- The incisal edge is almost at right angles to a line bisecting the crown labiolingually. This feature is characteristic of the tooth



MANDIBULAR CENTRAL INCISOR INCISAL VIEW

- When a line of vision is drawn (horizontally), more of the labial surface is seen than of the lingual surface
- Labiolingual dimension is larger than the mesiodistal



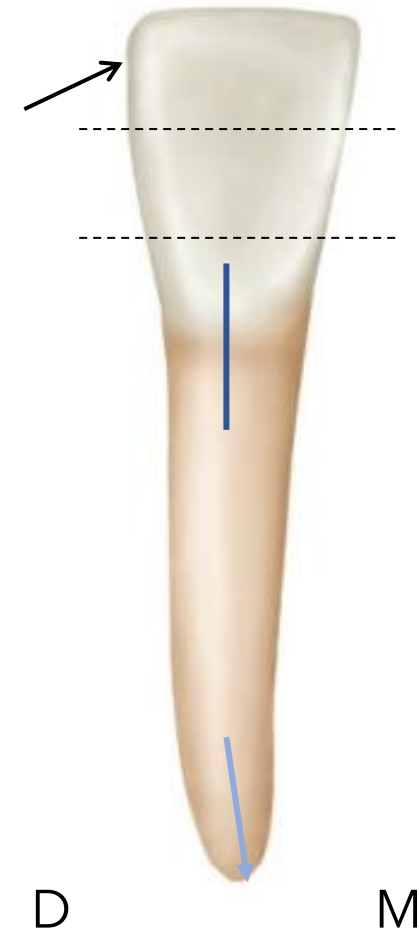
MANDIBULAR LATERAL INCISOR LABIAL VIEW

- The mandibular lateral incisor **is larger than the mandibular central incisor** (MD and BL) and appears rounded mesiodistally
- The incisal edge is straight, but it may appear slanted toward the distal
- The mesial incisal line angle sharp; the distal appears more rounded



MANDIBULAR LATERAL INCISOR LABIAL VIEW

- Proximal contacts are both within the incisal 1/3 of the crown
- More curvature in the distal crown outline; the crown is NOT considered symmetrical
- The crest of the gingival curvature of the cervical line: distal to the long axis
- The root tapers to a pointed apex. Usually straight



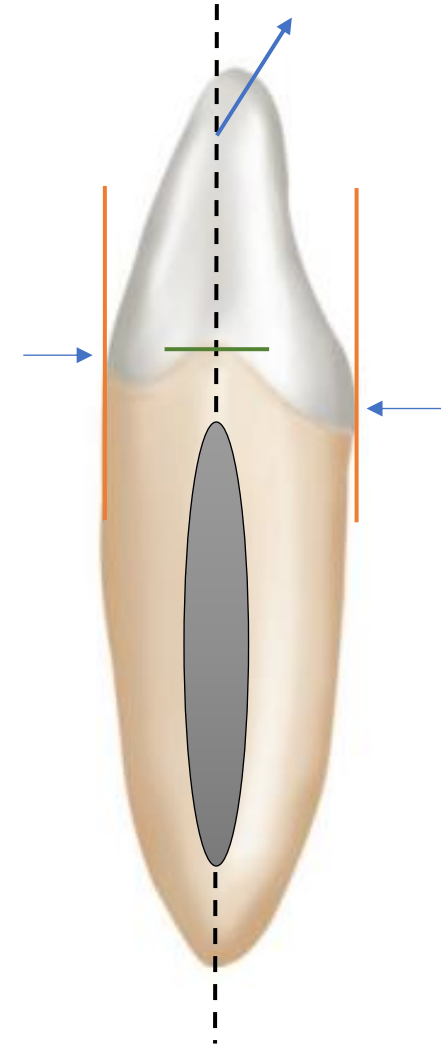
MANDIBULAR LATERAL INCISOR LINGUAL VIEW

- The lingual fossa is trapezoidal in shape, shallow and limited to incisal 1/3rd of the crown
- The cingulum is smoothly rounded, lies distal to the axis line of the root
- The marginal ridges are muted



MANDIBULAR LATERAL INCISOR MESIAL VIEW

- The incisal ridge will be lingual to the faciolingual long axis bisector of the tooth
- The crests of the cervical ridges (height of contour) will not be at the same level
- The mesial surface of the root will be similar in outline to that of the central incisor. A broad root depression will exist for most of the root length
- Curvature of cervical line less incisal



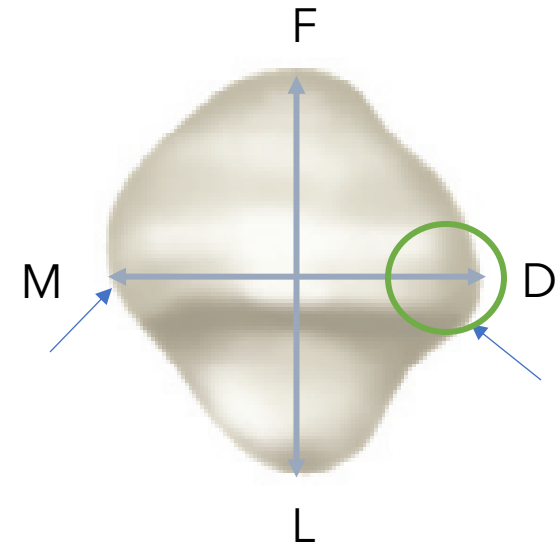
MANDIBULAR LATERAL INCISOR DISTAL VIEW

- The curvature of the cervical line is like mesial surface
- The root outline similar to that of the distal surface of mandibular central incisor
- Root depression maybe present
- The apex of the root will be in line with the faciolingual long axis bisector



MANDIBULAR LATERAL INCISOR INCISAL VIEW

- The labiolingual diameter is greater than the mesiodistal width
- The mesial contact area is centered faciolingually but the distal contact will be lingual of the mesiodistal line of vision
- The incisal edge is not at approximate right angles to a line bisecting labiolingually, the distal portion inclines lingually which gives the crown a **twisted or rotated appearance**



Distal portion of incisal ridge
is inclined lingually
(distolingual twist)

Mesial view of a mandibular left lateral incisor:
some of the lingual surface of the distal side is visible
because of the presence of **disto-lingual twist**.



COMPARISONS

FEATURE	MANDIBULAR CENTRAL INCISOR	MANDIBULAR LATERAL INCISOR
Size	Smaller in dimensions than lateral	Larger in dimension than central
Symmetry	The crown is bilaterally symmetrical (characteristic feature)	Not symmetrical. The crown appears rounded and inclined in the distal part
Labial aspect	Mesiodistally narrower than lateral (5mm)	Mesiodistally broader than central (5.5mm)
	Square and symmetrical appearance	Appears more rounded mesio-distally
	The mesioincisal angle is sharper and distoincisal is rounded	Both appear rounded, disto-incisal is more rounded than mesial
	The incisal ridge is straight	The incisal ridge slanted
	Root appears narrow, tapered and distally inclined in apical third	Root is straight and longer than central

COMAPARISONS

FEATURE	MANDIBULAR CENTRAL INCISOR	MANDIBULAR LATERAL INCISOR
Lingual aspect	Lingual fossa is trapezoidal and shallow	Lingual fossa is trapezoidal and shallow
	Cingulum placed centrally	Cingulum is distally offset
Proximal aspects	Curvature of cervical line on mesial aspect is more incisal	Curvature of cervical line is like mesial surface
	Narrow labiolingually than lateral (6mm)	Broader labiolingually than central (6.5mm)
	The contact points are on the incisal thirds	The mesial contact point is found on incisal third and distal is little cervical
	Root developmental depression is found on mesial side and developmental groove on distal side	Root developmental depressions are found on the mesial and distal sides

COMPARISONS

FEATURE	MANDIBULAR CENTRAL INCISOR	MANDIBULAR LATERAL INCISOR
Incisal aspect	The labiolingual diameter is greater than the mesiodistal width	The labiolingual diameter is greater than the mesiodistal width
	The incisal ridge is inclined lingually	The incisal ridge is inclined lingually
	The incisal aspect illustrates bilateral symmetry	The incisal aspect appears rounded
	The mesial half of crown is identical to the distal half	The distal portion of incisal ridge inclines lingually giving the crown twisted or rotated appearance (disto-lingual twist)



THANK YOU